

Ethan Mizuno

Houston, Texas | ethanmizuno16@gmail.com | 713-855-1095 | linkedin.com/in/ethanmizuno

EDUCATION

University of Washington | College of Engineering

Seattle, Washington

Bachelor of Science in Bioengineering | Minor in Data Science | GPA: 3.70

June 2024

- **Relevant Coursework:** Data Programming (Python), Computer Programming (Java), Data Visualization, Machine Learning, Computational Systems Biology, Systems and Control, Database Systems (SQL)
- **Awards/Honors:** Dean's List 10/12 Quarters
- **Leadership:** Fundraising Chair (Phi Delta Theta, Washington Alpha Chapter)

WORK EXPERIENCE

UW Medicine | Anesthesiology & Pain Medicine

Seattle, Washington

Full Stack Mobile App Developer (Capstone Project, Start Up Company)

January 2024 - Present

- Designed and developed a cross-platform mobile application for real-time monitoring of anesthesia data from multiple operating rooms using React Native and JavaScript. Application improved anesthesiology care team communication by 80%.
- Implemented machine learning algorithms for real-time data analysis, resulting in faster clinical responses by generating automated alerts for patient status changes during surgery.
- Collaborated with a multidisciplinary team, conducting weekly progress meetings with anesthesiology staff and stakeholders, ensuring seamless integration with hospital systems and on-time delivery of features.
- Optimized application performance and user experience for anesthesiologists by streamlining data visualization, improving operational efficiency across multiple operating rooms.

Predictive Sys-Bio Lab | UW Bioengineering

Seattle, Washington

Undergraduate Research Assistant

June 2023 - June 2024

- Engineered a comprehensive test suite using Python and the SBMLDiagrams API, ensuring robust software performance and reproducibility in biological simulations.
- Enhanced the reliability of SBMLDiagrams by identifying and resolving critical software issues through comprehensive testing and debugging, leading to improved user experience and performance.
- Increased cross-platform compatibility success rate by 20% by testing the SBMLDiagrams API with platforms like Copasi, demonstrating a strong understanding of cross-platform integration.

Vuori

Seattle, Washington

Sales Associate

August 2022 - June 2024

- Delivered exceptional customer service, consistently receiving positive feedback and contributing to a significant increase in customer satisfaction scores, helping drive average daily sales of \$25,000.
- Streamlined inventory management by efficiently restocking and organizing the inventory room, ensuring maximum accuracy and supporting seamless store operations.

PROJECTS

Predicting Clutch in Baseball

March 2024

- Utilized the Python package PyBaseball to web scrape statcast data from Baseball Savant and MLB Statcast
- Demonstrated data analysis with the Pandas library, generating matplotlib visualizations that illustrate trends in clutch versus non-clutch situations.

Analyzing Global Climate Change Data to Identify Key Environmental Indicators

March 2024

- Visualized 50+ years of climate data using the VegaLite API, enhancing comprehension of temperature anomalies and biodiversity trends.
- Analyzed global datasets from 100+ countries to reveal correlations between greenhouse gas emissions and biodiversity loss.

TECHNICAL SKILLS

Computer Languages: Python, Javascript, Java, R, MATLAB, SBML, HTML, CSS, SQL, Dart

Developer Tools: Git, VS Code, Jupyter, Rstudio, PyCharm

Libraries & Frameworks: React, Tableau, Tellurium, Flutter, Firebase, Pandas, NumPy, Matplotlib, TensorFlow